Additive Representations by Primitive Dyck Numbers: Exceptional Sets and Optimal Orders

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Abstract

Let $\mathcal{N}_{\mathrm{pD}}$ be the set consisting of 0 and the integers represented in binary by primitive Dyck words. For a positive even integer N, let $r(N)$ be the least number of positive elements of $\mathcal{N}_{\mathrm{pD}}$ whose sum is N; the associated sequence $a(n) = r(2n)$ is OEIS sequence A395858. We determine the complete exceptional set for six-summand representability: 46 alone satisfies $r(46) = 8$, the ten integers $34, 44, 98, 154, 198, 202, 206, 838, 842, 846$ satisfy $r(N) = 7$, and every other positive even integer satisfies $r(N) \leq 6$. Thus 848 is the sharp even threshold beyond which six summands always suffice. We also determine the exact relative asymptotic order. For every $k \geq 2$, one has $r(10 \cdot 4^{k+1} - 6) = 6$, so six summands are necessary infinitely often and the asymptotic order is exactly 6. The proof combines a regular sublanguage, finite interval certificates, a self-similar five-summand propagation, and a base-4 characterization of the halved primitive family by two-coloured Motzkin words. As a further consequence, that halved family is an asymptotic additive basis of order exactly 5.

Keywords: primitive Dyck number, additive representation, additive basis, asymptotic order, exceptional set, two-coloured Motzkin word, digital self-similarity, interval filling.
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1 Introduction

Every string of balanced parentheses can be written over the alphabet $\{1, 0\}$, with 1 for an opening bracket and 0 for a closing one, and so read as a binary integer. Which integers arise this way, and how additively rich is the set they form? We take up a natural variant of that question.

A *Dyck word* is a word $w \in \{1, 0\}^*$ with equally many 1's and 0's in which every prefix has at least as many 1's as 0's; thus 1 opens a parenthesis and 0 closes it. A nonempty Dyck word is *primitive* if none of its proper nonempty prefixes is itself balanced—equivalently, if the associated lattice path returns to height zero for the first time only at its endpoint. Such paths are also commonly called *irreducible* or *indecomposable* Dyck paths. Let $\mathcal{D}_{\mathrm{P}}$ denote the language of primitive Dyck words. Ordered by length and then lexicographically, its first eighteen elements are

$$10,\ 1100,\ 110100,\ 111000,\ 11010100,\ 11011000,$$
$$11100100,\ 11101000,\ 11110000,\ 1101010100,\ 1101011000,$$
$$1101100100,\ 1101101000,\ 1101110000,\ 1110010100,$$
$$1110011000,\ 1110100100,\ 1110101000.$$

For a word w over the base-m digits, let $[w]_m$ denote the integer it represents in base m. Adjoining 0 for convenience in additive representations, set

$$\mathcal{N}_{\mathrm{pD}} = \{0\} \cup \{[w]_2 : w \in \mathcal{D}_{\mathrm{P}}\}.$$

This convention does not affect the conclusions of the theorems below. The first eighteen positive elements are

$$2, 12, 52, 56, 212, 216, 228, 232, 240,$$
$$852, 856, 868, 872, 880, 916, 920, 932, 936;$$

this is OEIS sequence A057547 [**?**]. Every positive member of $\mathcal{N}_{\mathrm{pD}}$ is even, since every nonempty Dyck word ends in 0. Thus only even target integers can be represented by sums from $\mathcal{N}_{\mathrm{pD}}$.

The following sharper congruence observation will be used repeatedly.

Lemma 1. *The number* 2 *is the only positive primitive Dyck number congruent to* 2 *modulo* 4*. Every other positive primitive Dyck number is divisible by* 4*.*

Proof. Every nonempty Dyck word ends in 0. A primitive Dyck word of length at least four must in fact end in 00: if it ended in 10, the path would already have returned to height zero just before its last two letters. The word 10 represents 2, and every longer primitive word therefore represents a multiple of 4. □

Bell, Lidbetter, and Shallit studied the larger set obtained from *all* Dyck words, the *totally balanced numbers* (OEIS sequence A014486 [**?**]). Using regular underapproximations and automata, they proved that every even integer except

$$8,\ 18,\ 28,\ 38,\ 40,\ 82,\ 166$$

is a sum of at most three totally balanced numbers, while two summands do not suffice asymptotically [**?**, Thm. 14].

The primitive restriction is not a cosmetic variant of that result. Every nonempty Dyck word factors uniquely as a concatenation of primitive Dyck words, whereas here each summand must consist of a single factor. Consequently, a representation by unrestricted Dyck numbers does not in general yield a representation by the same number of primitive Dyck numbers.

Digit-defined additive bases have also been studied by automata-theoretic methods. Bell, Hare, and Shallit characterized automatic sets that form additive bases and gave an effective procedure for determining their orders [**?**]. Related decision procedures have yielded sharp additive results for binary palindromes [**?**] and binary squares [**?**]. The language of primitive

Dyck words is not regular, so those general finite-automaton methods do not directly settle the present problem.

Our proof differs at the structural point where regular approximation alone ceases to suffice. A regular sublanguage settles the congruence class 0 modulo 4, while the class 2 modulo 4 is handled by a self-similar closure property of the full primitive family after division by 4. The same family admits a base-4 description by two-coloured Motzkin words. Its self-similarity has two complementary effects: it propagates five-summand intervals, but its generation gaps also create an infinite sequence of obstructions to four-summand representations. These facts yield the complete exceptional set, the optimal uniform bound, the sharp eventual threshold, and the exact asymptotic order.

Within the Dyck words of length $2n$, the proportion that are primitive tends to $1/4$ as $n \to \infty$. Indeed, there are C_n Dyck words of length $2n$, and every primitive one is uniquely of the form $1u0$ with u a Dyck word of length $2n-2$; so there are C_{n-1} primitive words of length $2n$. From the Catalan formula $C_n = \frac{1}{n+1}\binom{2n}{n}$[1],

$$\frac{C_{n-1}}{C_n} = \frac{n+1}{2(2n-1)} \longrightarrow \frac{1}{4}.$$

Its arithmetic effect, however, is striking. Although all sufficiently large even integers need only three unrestricted Dyck summands, the primitive family has a small but nontrivial exceptional set. We show that precisely eleven positive even integers fail to be sums of at most six primitive Dyck numbers. Ten require seven summands, while 46 alone requires eight. All eleven exceptions lie below 848.

The need for eight summands is entirely local: below 48 the only positive primitive Dyck numbers are 2 and 12. The eventual six-summand bound, on the other hand, comes from two self-similar families. A regular subfamily handles multiples of 4, while the full primitive family after division by 4 handles the remaining congruence class.

For a positive even integer N, let $r(N)$ be the least number of positive primitive Dyck numbers whose sum is N, and define the central integer sequence of this paper by

$$a(n) = r(2n), \qquad n \geq 1.$$

This sequence has been registered in the OEIS as A395858 [**?**]. Its first fifty terms are

$$1, 2, 3, 4, 5, 1, 2, 3, 4, 5, 6, 2, 3, 4, 5, 6, 7, 3, 4, 5, 6, 7, 8, 4, 5,$$
$$1, 2, 1, 2, 3, 4, 2, 3, 2, 3, 4, 5, 3, 4, 3, 4, 5, 6, 4, 5, 4, 5, 6, 7, 5.$$

The following Python program reproduces the displayed terms and generates 50,000 terms of the sequence, making some recurring patterns visible: `A395858.py` at the fixed repository snapshot `3525cfb`.

Theorem 2. *The values of $r(N)$ exceeding six are exactly as follows:*

$$r(46) = 8,$$
$$r(N) = 7 \iff N \in \{34, 44, 98, 154, 198, 202, 206, 838, 842, 846\}.$$

[1]The grammar $S \to \varepsilon \mid 1S0S$ gives the unique decomposition $1u0v$ of every nonempty Dyck word, where u and v are Dyck words. Hence $C_0 = 1$ and $C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}$. If $B_n = \frac{1}{n+1}\binom{2n}{n}$, then the binomial convolution identity gives $B_0 = 1$ and $B_n = \sum_{k=0}^{n-1} B_k B_{n-1-k}$. Therefore induction yields $C_n = B_n$.

Thus every other positive even integer satisfies $r(N) \le 6$. In other words, every even integer $N \ge 848$ is a sum of at most six primitive Dyck numbers.

Equivalently, for the OEIS sequence A395858 [**?**], defined by $a(n) = r(2n)$,

$$\max_{n\ge 1} a(n) = 8, \qquad a(n) = 8 \iff n = 23,$$

and

$$a(n) = 7 \iff n \in \{17, 22, 49, 77, 99, 101, 103, 419, 421, 423\}.$$

Thus the main theorem gives a complete determination of all terms of $a(n)$ exceeding six, rather than only an eventual estimate.

The eventual bound six is also best possible.

Theorem 3. *Let*

$$h_{\mathrm{asym}} = \min\{h : \textit{every sufficiently large even } N \textit{ satisfies } r(N) \le h\}.$$

Then $h_{\mathrm{asym}} = 6$. More precisely, for every integer $k \ge 2$,

$$N_k = 10 \cdot 4^{k+1} - 6$$

satisfies $r(N_k) = 6$. Equivalently,

$$a(5 \cdot 4^{k+1} - 3) = 6 \qquad (k \ge 2).$$

Thus the finite exceptional set above coexists with infinitely many large integers for which the six-summand upper bound is attained.

The exceptional value 46 already explains why the uniform bound can never be smaller than eight.

Proposition 4. *The integer 46 requires exactly eight positive primitive Dyck numbers: it is a sum of eight, but it is not a sum of at most seven.*

Proof. Below 48 the only positive primitive Dyck numbers are 2 and 12. Indeed, the primitive words of length two and four are 10 and 1100, whereas every primitive word of length at least six begins with 11 and therefore represents an integer at least $[110000]_2 = 48$. Thus any representation of 46 has the form

$$46 = 12a + 2b, \qquad a, b \ge 0.$$

Equivalently, $6a + b = 23$. Since $12 \cdot 4 > 46$, we have $a \le 3$, and hence the number of summands satisfies

$$a + b = 23 - 5a \ge 8.$$

Equality is attained by

$$46 = 12 + 12 + 12 + 2 + 2 + 2 + 2 + 2.$$

□

For example,

$$2026 = 2 + 240 + 852 + 932 = 2 + 216 + 872 + 936.$$

The seven distinct integers appearing in these two representations all have primitive Dyck binary expansions.

The sumset notation used below is standard in additive-basis theory [**?**, Chap. 1]. For a nonnegative integer k, let kX denote the set of sums of k not necessarily distinct members of X, with $0X = \{0\}$, and put

$$F_k(X) = \bigcup_{j=0}^{k} jX.$$

Thus $F_k(X)$ is the set of sums of at most k elements of X. For a scalar c, write $c \cdot X = \{cx : x \in X\}$ for the dilation of X.

The mod-4 property motivates the central halved family

$$\left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right) = \{v/4 : v \in \mathcal{N}_{\mathrm{pD}},\ v \geq 12\}, \tag{1}$$

so that

$$\mathcal{N}_{\mathrm{pD}} = \{0, 2\} \cup 4\left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right). \tag{2}$$

The first elements of $\left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$ are

$$3, 13, 14, 53, 54, 57, 58, 60, 213, 214, 217, \ldots.$$

2 The base-4 structure of $\left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$

Assign weights to the base-4 digits by

$$\sigma(3) = 1, \qquad \sigma(0) = -1, \qquad \sigma(1) = \sigma(2) = 0.$$

A word $w = c_0 \cdots c_{k-1} \in \{0, 1, 2, 3\}^k$ is called a *two-coloured Motzkin word* if every prefix has nonnegative total weight and the total weight of w is 0. The two zero-weight digits 1 and 2 are regarded as distinct colours. Write $[w]_4$ for the value of w as a base-4 numeral, with $[\varepsilon]_4 = 0$. For a positive integer n, let $\mathrm{bin}(n)$ and $\mathrm{quad}(n)$ denote its standard binary and base-4 expansions, respectively.

Lemma 5 (Base-4 characterization)**.** *A positive integer p lies in $\left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$ if and only if its base-4 expansion has the form $3w$, where w is a two-coloured Motzkin word.*

Proof. Suppose first that $p \in \left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$. Then

$$v := 4p \in \mathcal{N}_{\mathrm{pD}}, \qquad v \geq 12.$$

Write the binary expansion of v as

$$b_1 b_2 \cdots b_{2L} \qquad (b_i \in \{0, 1\}),$$

and define its height function by

$$h(t) := \sum_{i=1}^{t}(2b_i - 1) \qquad (0 \le t \le 2L).$$

Since v is represented by a primitive Dyck word,

$$h(0) = h(2L) = 0, \qquad h(t) > 0 \quad (1 \le t < 2L).$$

Group the binary digits into consecutive pairs and put

$$d_j := 2b_{2j-1} + b_{2j} \in \{0, 1, 2, 3\} \qquad (1 \le j \le L).$$

Then $d_1 \cdots d_L$ is the base-4 expansion of v. The correspondence

$$11 \leftrightarrow 3, \quad 10 \leftrightarrow 2, \quad 01 \leftrightarrow 1, \quad 00 \leftrightarrow 0$$

shows that

$$h(2j) - h(2j-2) = 2\sigma(d_j),$$

and hence

$$h(2j) = 2\sum_{i=1}^{j}\sigma(d_i) \qquad (0 \le j \le L). \tag{3}$$

Because the primitive word has length at least four, its first pair is 11 and its last pair is 00: otherwise it would return to height 0 after its first two bits or before its last two bits. Thus

$$d_1 = 3, \qquad d_L = 0.$$

For $1 \le j < L$, the integer $h(2j)$ is a positive even number, so

$$\sum_{i=1}^{j}\sigma(d_i) = \frac{h(2j)}{2} \ge 1.$$

Dividing $v = 4p$ by 4 deletes the final base-4 digit 0. Therefore the base-4 expansion of p is $3w$, where

$$w := d_2 d_3 \cdots d_{L-1}.$$

For every prefix of w of length r, with $0 \le r \le L-2$, one has

$$\sum_{i=2}^{r+1}\sigma(d_i) = \sum_{i=1}^{r+1}\sigma(d_i) - \sigma(d_1) \ge 1 - 1 = 0.$$

Moreover, by (**??**),

$$\sum_{i=1}^{L}\sigma(d_i) = \frac{h(2L)}{2} = 0.$$

Since $\sigma(d_1) = 1$ and $\sigma(d_L) = -1$, it follows that

$$\sum_{i=2}^{L-1} \sigma(d_i) = 0.$$

Thus w is a two-coloured Motzkin word.

Conversely, suppose that

$$\mathrm{quad}(p) = 3w, \qquad w = e_1 \cdots e_m$$

is a two-coloured Motzkin word. Thus

$$\sum_{i=1}^{r} \sigma(e_i) \geq 0 \quad (0 \leq r \leq m), \qquad \sum_{i=1}^{m} \sigma(e_i) = 0.$$

Append a final base-4 digit 0 and expand the digits of $3w0$ by the block map

$$\phi(3) = 11, \qquad \phi(2) = 10, \qquad \phi(1) = 01, \qquad \phi(0) = 00.$$

Let $B = \phi(3w0)$ and let h be its binary height function. At the pair boundaries,

$$h(2) = 2$$

and, for $0 \leq r \leq m$,

$$h(2r+2) = 2\left(1 + \sum_{i=1}^{r} \sigma(e_i)\right) \geq 2.$$

After the appended final digit 0 has been read,

$$h(2m+4) = 2\left(1 + \sum_{i=1}^{m} \sigma(e_i) - 1\right) = 0.$$

It remains to check positions inside the two-bit blocks. The blocks corresponding to 3 and 2 begin with 1, so the height first increases. A block corresponding to 1 is 01; its initial height is at least 2, so after the first bit its height is still at least 1. If a digit 0 occurs inside w, then the Motzkin prefix condition implies that the weight before that digit is at least 1. Hence the binary height before its block is at least 4, and after the first 0 it is at least 3. Finally, the appended terminal block 00 begins at height 2, passes through height 1, and ends at height 0.

Consequently B stays strictly above height 0 before its final bit and returns to height 0 at the end. It is therefore a primitive Dyck word. Its base-4 digits are $3w0$, so it represents $4p$. Also $p \geq 3$, and hence $4p \geq 12$. Thus $4p \in \mathcal{N}_{\mathrm{pD}}$ and $p \in \left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$. $\square$

The characterization gives the self-similarity used later.

Corollary 6. *If $p \in \left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$, then $4p+1$ and $4p+2$ also belong to $\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$.*

Proof. Write

$$\mathrm{quad}(p) = 3w,$$

where w is a two-coloured Motzkin word. Then

$$\mathrm{quad}(4p+1) = 3w1, \qquad \mathrm{quad}(4p+2) = 3w2.$$

The digits 1 and 2 both have σ-weight 0, so appending either of them preserves the two-coloured Motzkin condition. The claim follows from Lemma **??**. $\square$

We next determine the smallest and largest values in each generation.

Lemma 7 (Extremal two-coloured Motzkin values)**.** *For every $\ell \geq 0$ and every two-coloured Motzkin word w of length ℓ,*

$$\frac{4^\ell - 1}{3} \leq [w]_4 \leq 4^\ell - 2^\ell. \tag{4}$$

Both bounds are attained. The lower bound is attained by 1^ℓ, and the upper bound is attained by

$$w_{\max,\ell} = \begin{cases} 3^a 0^a, & \ell = 2a, \\ 3^a 2\, 0^a, & \ell = 2a+1. \end{cases}$$

Proof. We use induction on ℓ. The assertion is immediate for $\ell = 0$. Write

$$w = c_0 c_1 \cdots c_{\ell-1}, \qquad c_j \in \{0,1,2,3\}.$$

Lower bound. Since $\sigma(0) = -1$, the first digit cannot be 0. If $c_0 \in \{1,2\}$, then the suffix $w' = c_1 \cdots c_{\ell-1}$ is again a two-coloured Motzkin word. Hence

$$[w]_4 \geq 4^{\ell-1} + [w']_4 \geq 4^{\ell-1} + \frac{4^{\ell-1}-1}{3} = \frac{4^\ell - 1}{3}.$$

If $c_0 = 3$, then

$$[w]_4 \geq [3\,\underbrace{00\cdots0}_{\ell-1 \text{ digits}}]_4 = 3 \cdot 4^{\ell-1} \geq \frac{4^\ell-1}{3}.$$

Equality is attained by $w = 1^\ell$.

Upper bound. If $c_0 \in \{1,2\}$, then w' is again a two-coloured Motzkin word, so

$$\begin{aligned} [w]_4 &= [c_0 c_1 \cdots c_{\ell-1}]_4 \\ &\leq [2c_1 \cdots c_{\ell-1}]_4 \\ &\leq 2 \cdot 4^{\ell-1} + \left(4^{\ell-1} - 2^{\ell-1}\right) \\ &= 3 \cdot 4^{\ell-1} - 2^{\ell-1} \\ &\leq 4^\ell - 2^\ell. \end{aligned}$$

Now suppose $c_0 = 3$, and let r be the length of the shortest nonempty prefix of w having total weight 0. Then $2 \leq r \leq \ell$, $c_{r-1} = 0$, and

$$u = c_1 \cdots c_{r-2}, \qquad v = c_r \cdots c_{\ell-1}$$

are two-coloured Motzkin words of lengths $r-2$ and $\ell-r$, respectively. Thus $w = 3u0v$. By place value and the induction hypothesis,

$$\begin{aligned}[w]_4 &= 3\cdot 4^{\ell-1} + [u]_4\, 4^{\ell-r+1} + [v]_4 \\ &\le 3\cdot 4^{\ell-1} + \bigl(4^{r-2} - 2^{r-2}\bigr)4^{\ell-r+1} + 4^{\ell-r} - 2^{\ell-r} \\ &= 4^{\ell} - 2^{2\ell-r} + 2^{2\ell-2r} - 2^{\ell-r} \\ &= 4^{\ell} - \bigl(2^{2\ell-r} - 2^{2\ell-2r}\bigr) - 2^{\ell-r}.\end{aligned}$$

It remains to verify

$$2^{2\ell-r} - 2^{2\ell-2r} \ge 2^{\ell} - 2^{\ell-r}.$$

After factoring, this is

$$2^{2\ell-2r}(2^r-1) \ge 2^{\ell-r}(2^r-1),$$

which follows from $r \le \ell$. This proves the upper bound.

Finally, the displayed words $w_{\max,\ell}$ are two-coloured Motzkin words. A geometric-series calculation gives

$$[w_{\max,\ell}]_4 = 4^{\ell} - 2^{\ell},$$

so the upper bound is attained. □

An element of $\bigl(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD},\ge 12}\bigr)$ belongs to *generation* d if its base-4 expansion has exactly d digits. By Lemma **??**, a generation-d element has value $3\cdot 4^{d-1} + [w]_4$, where w has length $d-1$.

Corollary 8 (Generation bounds)**.** *For every $d \ge 1$, every generation-d element of $\bigl(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD},\ge 12}\bigr)$ lies in*

$$[g_d, U_d] = \left[3\cdot 4^{d-1} + \frac{4^{d-1}-1}{3},\, 4^d - 2^{d-1}\right], \tag{5}$$

and both endpoints are attained. In particular,

(a) *there is no element of $\bigl(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD},\ge 12}\bigr)$ in $(U_d, 3\cdot 4^d)$;*

(b) *for every $k \ge 0$,*

$$3g_{k+1} = 10\cdot 4^k - 1. \tag{6}$$

Proof. The interval follows from Lemmas **??** and **??**. The attainment statements follow from the extremal words in Lemma **??**. Part (a) follows because the next generation begins at $g_{d+1} > 3\cdot 4^d$. Finally,

$$3g_{k+1} = 3\left(3\cdot 4^k + \frac{4^k-1}{3}\right) = 10\cdot 4^k - 1.$$

□

3 A regular family and six-term interval filling

Inside the base-4 family described in Section **??**, a simple regular pattern runs through the primitive Dyck words. For example, the binary expansion of 868 factors as

$$\mathrm{bin}(868) = 1101100100 = 11\,01\,10\,01\,00 \in 11(01 \mid 10)^*00.$$

Pairing the binary digits turns the initial block 11 into the base-4 digit 3, the final block 00 into 0, and each middle block 01 or 10 into 1 or 2. This suggests that sums of a fixed number of such base-4 integers might fill long intervals. The mixed sumset lemma below makes the idea precise.

Let

$$\mathcal{E} = \{0\} \cup \Big\{[3\varepsilon_1 \cdots \varepsilon_k]_4 : k \geq 0,\ \varepsilon_i \in \{1,2\}\Big\}.$$

When $k = 0$ the string $\varepsilon_1 \cdots \varepsilon_k$ is empty, so the corresponding element is $[3]_4 = 3$. Multiplying by 4 appends a base-4 zero. Replacing each base-4 digit by its two binary digits,

$$\begin{aligned} 3_{(4)} &\longleftrightarrow 11_{(2)}, & 1_{(4)} &\longleftrightarrow 01_{(2)}, \\ 2_{(4)} &\longleftrightarrow 10_{(2)}, & 0_{(4)} &\longleftrightarrow 00_{(2)}. \end{aligned}$$

we find that every positive element of $4 \cdot \mathcal{E}$ has binary expansion in $11(01 \mid 10)^*00$. After the initial 11 the path stands at height 2; each middle block 01 or 10 returns it to height 2; and the final 00 brings it to height zero for the first time. Thus every positive element of $4 \cdot \mathcal{E}$ is a primitive Dyck number, and, since $0 \in \mathcal{N}_{\mathrm{pD}}$,

$$4 \cdot \mathcal{E} \subseteq \mathcal{N}_{\mathrm{pD}}. \tag{7}$$

The crux is that six elements of $\mathcal{E}$ already cover every integer from 25 on.

Proposition 9. *Every integer $n \geq 25$ belongs to $6\mathcal{E}$.*

We prove this by a finite-digit interval argument. The role of the auxiliary sets below is worth explaining first. When an m-digit element of $\mathcal{E}_m$ is split into its leading digit and its lower digits, the lower $m-1$ digits form a base-4 correction whose digits lie in $\{0,1\}$. The set Δ_m records exactly these corrections. When some summands in a six-term sum use a new leading digit, the remaining lower-digit problem becomes a mixed sum of copies of Δ_{m-1} and $\mathcal{E}_{m-1}$. The mixed sumsets $S_{t,m}$ are designed to keep track of all these possibilities simultaneously.

Set $u_0 = 0$ and, for $m \geq 1$,

$$u_m = [\underbrace{11\cdots1}_{m \text{ digits}}]_4 = 1 + 4 + \cdots + 4^{m-1} = \frac{4^m - 1}{3}, \tag{8}$$

so that

$$\begin{aligned} 4^{m-1} &= 3u_{m-1} + 1 \quad (m \geq 1), \\ u_{m-1} &= 4u_{m-2} + 1 \quad (m \geq 2). \end{aligned} \tag{9}$$

Define the correction set

$$\Delta_m = \Big\{[\delta_{m-1} \cdots \delta_0]_4 : \delta_j \in \{0,1\}\Big\} = \Big\{\textstyle\sum_{j=0}^{m-1} \delta_j 4^j : \delta_j \in \{0,1\}\Big\},$$

so that $\max \Delta_m = u_m$. Let $\mathcal{E}_m$ be the finite subset of $\mathcal{E}$ consisting of 0 together with the elements having at most m base-4 digits. For example,

$$\Delta_2 = \{[00]_4, [01]_4, [10]_4, [11]_4\} = \{0, 1, 4, 5\},$$
$$\mathcal{E}_2 = \{0, [3]_4, [31]_4, [32]_4\} = \{0, 3, 13, 14\}.$$

For $0 \le t \le 6$ set

$$S_{t,m} = t\Delta_m + (6-t)\mathcal{E}_m. \tag{10}$$

The largest element of $\mathcal{E}_m$ is

$$\max \mathcal{E}_m = [3\ \underbrace{22\cdots 2}_{m-1 \text{ digits}}]_4 = 3 \cdot 4^{m-1} + 2u_{m-1},$$

and for $m \ge 2$ equation (**??**) gives

$$\max \mathcal{E}_{m-1} = 11u_{m-2} + 3 = \frac{11u_{m-1} + 1}{4}. \tag{11}$$

Since $\max \Delta_m = u_m$,

$$\max S_{t,m} = tu_m + (6-t) \max \mathcal{E}_m. \tag{12}$$

Moreover $\max \mathcal{E}_m > u_m$, so $\max S_{t,m}$ decreases with t. Throughout, $[a, b]$ denotes the integer interval $\{a, a+1, \ldots, b\}$.

Lemma 10. *For every $m \ge 2$ and every $2 \le t \le 6$,*

$$S_{t,m} = [0, \max S_{t,m}], \tag{13}$$

whereas

$$S_{1,m} = [0, \max S_{1,m}] \setminus \{2\}. \tag{14}$$

Moreover,

$$[25, \ell_m] \subseteq 6\mathcal{E}_m, \qquad \ell_m = \frac{33}{8} 4^m - 4. \tag{15}$$

Proof. We first prove (**??**) and (**??**) in the base case $m = 2$. Recall that

$$\Delta_2 = \{0, 1, 4, 5\}, \qquad \mathcal{E}_2 = \{0, 3, 13, 14\}.$$

Direct calculation gives $2\Delta_2 = \{0, 1, 2, 4, 5, 6, 8, 9, 10\}$, and adding one more copy of Δ_2 fills every remaining gap, so $3\Delta_2 = [0, 15]$ and hence

$$t\Delta_2 = [0, 5t] \qquad (3 \le t \le 6).$$

On the other hand,

$$2\mathcal{E}_2 = \{0, 3, 6, 13, 14, 16, 17, 26, 27, 28\},$$

$$\begin{aligned} 3\mathcal{E}_2 = \{0, 3, 6, 9\} &\cup [13, 14] \cup [16, 17] \cup [19, 20] \\ &\cup [26, 31] \cup [39, 42], \end{aligned}$$

$$\begin{aligned} 4\mathcal{E}_2 = \{0, 3, 6, 9\} &\cup [12, 14] \cup [16, 17] \cup [19, 20] \\ &\cup [22, 23] \cup [26, 34] \cup [39, 45] \cup [52, 56], \end{aligned}$$

$$\begin{aligned} 5\mathcal{E}_2 = \{0, 3, 6, 9\} &\cup [12, 17] \cup [19, 20] \cup [22, 23] \\ &\cup [25, 37] \cup [39, 48] \cup [52, 59] \cup [65, 70]. \end{aligned}$$

Adding Δ_2 to $5\mathcal{E}_2$, and $2\Delta_2$ to $4\mathcal{E}_2$, gives

$$S_{1,2} = \{0,1\} \cup [3,75], \qquad S_{2,2} = [0,66].$$

Similarly,

$$\begin{aligned}
S_{3,2} &= 3\Delta_2 + 3\mathcal{E}_2 = [0,15] + 3\mathcal{E}_2 = [0,57],\\
S_{4,2} &= 4\Delta_2 + 2\mathcal{E}_2 = [0,20] + 2\mathcal{E}_2 = [0,48],\\
S_{5,2} &= 5\Delta_2 + \mathcal{E}_2 = [0,25] + \mathcal{E}_2 = [0,39],\\
S_{6,2} &= 6\Delta_2 = [0,30].
\end{aligned}$$

Collecting these,

t	$S_{t,2}$
1	$[0,75] \setminus \{2\}$
2	$[0,66]$
3	$[0,57]$
4	$[0,48]$
5	$[0,39]$
6	$[0,30]$

which proves (**??**) and (**??**) for $m = 2$. For (**??**) we must check $[25, \ell_2] = [25, 62] \subseteq 6\mathcal{E}_2$. The intervals $[26, 28]$, $[39, 42]$, and $[52, 56]$ are the sums of two, three, and four members of $\{13, 14\}$, respectively, so

$$\begin{aligned}
25 &= 13 + 3 + 3 + 3 + 3 + 0,\\
[26,40] &= \{26,27,28\} + 3\cdot\{0,1,2,3,4\},\\
[39,51] &= [39,42] + 3\cdot\{0,1,2,3\},\\
[52,62] &= [52,56] + 3\cdot\{0,1,2\}.
\end{aligned}$$

This completes the base case. The lower endpoint is sharp: $24 \notin 6\mathcal{E}$, since the only positive elements of $\mathcal{E}$ below 25 are $3, 13$, and 14, and no sum of at most six of these is 24.

Now let $m \geq 3$ and assume that (**??**)–(**??**) hold at level $m-1$. To lighten the notation during this step, set

$$u = u_{m-1}, \qquad q = q_m = [3\,\underbrace{11\cdots1}_{m-1\text{ digits}}]_4 = 3\cdot 4^{m-1} + u, \qquad M = \max \mathcal{E}_{m-1}.$$

Then

$$4^{m-1} = 3u + 1, \qquad M = \frac{11u+1}{4}.$$

Moreover,

$$q = 10u + 3. \tag{16}$$

We shall repeatedly use

$$2\cdot 4^{m-1} + 4u = q - 1, \tag{17}$$

whose two sides are both equal to $10u + 2$. Splitting according to the leading base-4 digit gives

$$\Delta_m = \Delta_{m-1} \cup (4^{m-1} + \Delta_{m-1}), \qquad \mathcal{E}_m = \mathcal{E}_{m-1} \cup (q + \Delta_{m-1}).$$

Indeed, every new m-digit element of Δ_m has leading digit 1, whereas a new m-digit element of $\mathcal{E}_m$ has leading digit 3 followed by a lower $(m-1)$-digit correction from Δ_{m-1}. If s of the t copies of Δ_m use their shifted part and v of the $6-t$ copies of $\mathcal{E}_m$ use their new m-digit part, then the sumset in (**??**) decomposes as

$$S_{t,m} = \bigcup_{v=0}^{6-t} \bigcup_{s=0}^{t} \bigl(s \cdot 4^{m-1} + vq + S_{t+v,m-1}\bigr). \tag{18}$$

Thus, for fixed v, the parameter s produces $t+1$ translates by multiples of 4^{m-1}; increasing v then moves to the next large block:

$$\underbrace{[vq,\ \cdots]}_{v\text{-block}} \qquad \underbrace{[(v+1)q,\ \cdots]}_{(v+1)\text{-block}}.$$

The rest of the proof checks that neighboring translates and neighboring v-blocks overlap.

We first treat $t \geq 2$. The induction hypothesis (**??**) gives $S_{t+v,m-1} = [0, \max S_{t+v,m-1}]$ for every value of v in (**??**). For fixed v, consecutive values of s shift this interval by 4^{m-1}. Since $\max S_{j,m-1}$ decreases with j,

$$\max S_{t+v,m-1} \geq \max S_{6,m-1} = 6u \geq 3u = 4^{m-1} - 1,$$

so these $t+1$ translates overlap and form

$$[vq,\ vq + t \cdot 4^{m-1} + \max S_{t+v,m-1}].$$

The blocks for consecutive values of v also overlap. Whenever a next block exists we have $t + v \leq 5$, and since $t \geq 2$,

$$\begin{aligned} t \cdot 4^{m-1} + \max S_{t+v,m-1} &\geq 2 \cdot 4^{m-1} + \max S_{5,m-1} \\ &= 2 \cdot 4^{m-1} + 5u + M \\ &= (2 \cdot 4^{m-1} + 4u) + (u + M) \\ &\geq q - 1, \end{aligned}$$

by (**??**). Hence the union in (**??**) is the full integer interval from 0 to its largest element $\max S_{t,m}$. This proves (**??**) at level m.

For $t = 1$ the first block retains the single missing integer 2. For each fixed $v \geq 1$, the two s-translates involve $S_{1+v,m-1}$ with $1 + v \geq 2$; they overlap because

$$\max S_{1+v,m-1} \geq \max S_{6,m-1} = 6u \geq 4^{m-1} - 1.$$

Every later v-block is therefore an interval. Since $\max S_{j,m-1}$ decreases with j, it suffices to check the last pair of consecutive v-blocks. We have

$$\begin{aligned} 4^{m-1} + \max S_{5,m-1} - (q-1) &= M - 2u - 1 \\ &= \frac{11u+1}{4} - 2u - 1 = \frac{3u-3}{4} \geq 0, \end{aligned}$$

so consecutive v-blocks overlap. Finally,

$$\begin{aligned}\max S_{1,m-1} - (4^{m-1}+2) &= u + 5M - 4^{m-1} - 2\\ &= \frac{47u-7}{4} \geq 0.\end{aligned}$$

Thus, within the first v-block, the unshifted long interval reaches $4^{m-1}+2$ and joins the next s-translate without a gap. Hence

$$S_{1,m} = [0, \max S_{1,m}] \setminus \{2\},$$

which proves (**??**) at level m.

It remains to propagate the interval inside $6\mathcal{E}_m$. Taking $t=0$ in (**??**),

$$6\mathcal{E}_m = \bigcup_{v=0}^{6} \big(vq + S_{v,m-1}\big).$$

The $v=0$ block is $S_{0,m-1} = 6\mathcal{E}_{m-1}$, which contains $[25, \ell_{m-1}]$ by the induction hypothesis. Since $m \geq 3$ we have $u \geq 5$, and

$$\ell_{m-1} - (q+2) = \frac{19u-39}{8} \geq 0.$$

The $v=1$ block, namely $q + S_{1,m-1}$, contains $\{q, q+1\} \cup [q+3, q+\max S_{1,m-1}]$ by the induction hypothesis. Because the preceding interval reaches through $q+2$, the two join without a gap. To join the $v=j$ block to the $v=j+1$ block for $j=1,2,3$, it is enough to have

$$\max S_{j,m-1} \geq q-1.$$

Since $\max S_{j,m-1}$ decreases with j, the most restrictive case is $j=3$; hence it suffices to check

$$\begin{aligned}\max S_{3,m-1} - (q-1) &= 3u + 3M - (10u+2)\\ &= \frac{5u-5}{4} \geq 0.\end{aligned}$$

Thus $\max S_{1,m-1},\ \max S_{2,m-1},\ \max S_{3,m-1} \geq q-1$, and the blocks $v=1,2,3,4$ join in turn and cover through

$$4q + \max S_{4,m-1} = \frac{33}{8}4^m - 4 = \ell_m.$$

The remaining blocks $v=5,6$ may begin after a gap, but they are not needed for the interval propagated here. This establishes (**??**)–(**??**) at level m and completes the induction. □

Since $\ell_m \to \infty$ and $\mathcal{E} = \bigcup_{m\geq 1}\mathcal{E}_m$, Proposition **??** follows at once.

The preceding proposition handles the congruence class 0 modulo 4: every multiple of 4 at least 100 is a sum of at most six primitive Dyck numbers.

4 A five-summand interval for the full halved family

The regular family $\mathcal{E}$ is sufficient for multiples of 4, but it does not by itself provide the five-summand interval needed for the other congruence class. We now use the full family $\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$ from (**??**), together with its self-similar closure in Corollary **??**.

We next establish a five-summand interval for $\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$. The finite base uses the two subsets

$$L = \{3, 13, 14, 53, 54, 57, 58, 60\} \subseteq \left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right). \tag{19}$$

and

$$\begin{aligned} H = \{&213, 214, 217, 218, 220, 229, 230, 233, 234, 236, \\ &241, 242, 244, 248\} \subseteq \left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right). \end{aligned} \tag{20}$$

The following tables display all the corresponding primitive Dyck numbers. In each entry the binary word $[4h]_2$ is primitive, and hence $h \in \left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$.

$4L$

h	$[4h]_2$	h	$[4h]_2$
3	1100	53	11010100
13	110100	54	11011000
14	111000	57	11100100
58	11101000	60	11110000

$4H$

h	$[4h]_2$	h	$[4h]_2$
213	1101010100	233	1110100100
214	1101011000	234	1110101000
217	1101100100	236	1110110000
218	1101101000	241	1111000100
220	1101110000	242	1111001000
229	1110010100	244	1111010000
230	1110011000	248	1111100000

The finite computations can be reproduced using the Python program `verify_certificates_for_paper.py`, available as a fixed repository copy at commit `3525cfb`. The archived version passed the repository's Python CI check at that commit (Python CI run #1). The same program is included as a supplementary file, together with a README giving the execution command and the expected output.

The five statements in Lemma **??** are inclusion certificates: the program checks that each displayed interval is contained in the corresponding sumset, and does not claim that the sumset equals that interval. By contrast, for Lemma **??** the program computes the relevant sets truncated to $[0, 211]$ exactly and checks equality with the displayed unions of intervals.

Lemma 11 (Finite interval certificate)**.** *The following inclusions hold:*

$$\begin{array}{c|c} \textit{sumset} & \textit{integer interval contained} \\ \hline 5L & [215,254] \\ H+4L & [245,486] \\ 2H+3L & [439,674] \\ 3H+2L & [645,862] \\ 4H+L & [855,1046]. \end{array} \tag{21}$$

Proof. For the first inclusion in (**??**), namely $[215, 254] \subseteq 5L$, partition the set L defined in (**??**) as

$$A = \{53, 54, 57, 58, 60\}, \qquad B = \{3, 13, 14\},$$

so that $L = A \sqcup B$. Successive additions give

$$2A = [106, 108] \cup [110, 118] \cup \{120\},$$

and

$$3A = [159, 178] \cup \{180\}, \qquad 4A = [212, 238] \cup \{240\}.$$

Hence $4A + B = [215, 254]$, proving the first inclusion in (**??**).

For the second inclusion in (**??**), namely $[245, 486] \subseteq H + 4L$, repeated addition of L gives

$$\begin{aligned} 4L \supseteq\ & [32, 34] \cup [42, 45] \cup [52, 56] \cup [82, 102] \\ & \cup [116, 148] \cup [162, 194] \cup [212, 238]. \end{aligned} \tag{22}$$

Among the seven intervals on the right-hand side of (**??**), adding the set H defined in (**??**) to the first three short intervals gives

$$\begin{aligned} H + [32, 34] &\supseteq [245, 254] \cup [261, 270] \cup [273, 278] \cup [280, 282], \\ H + [42, 45] &\supseteq [255, 265] \cup [271, 281] \cup [283, 293], \\ H + [52, 56] &\supseteq [265, 276] \cup [281, 304]. \end{aligned}$$

Their union contains $[245, 304]$. Since $\min H = 213$, $\max H = 248$, and consecutive elements of H differ by at most 9, if $I = [a, b]$ satisfies $b - a \geq 8$, then

$$H + I = [213 + a, 248 + b].$$

Applying this to the final four intervals in (**??**) gives, respectively,

$$[295, 350], \quad [329, 396], \quad [375, 442], \quad [425, 486].$$

These intervals overlap, completing the proof that $[245, 486] \subseteq H + 4L$.

For the third inclusion in (**??**), namely $[439, 674] \subseteq 2H + 3L$, the following contained intervals suffice:

$$2H \supseteq [426, 428] \cup [430, 438] \cup [446, 478] \cup \{496\},$$

and

$$3L \supseteq \{9\} \cup [19,20] \cup [39,42] \cup [73,77] \cup [109,111] \\ \cup [113,134] \cup [159,178] \cup \{180\}.$$

The first two components of $2H$ give

$$[430,438] + 9 = [439,447], \quad [426,428] + [19,20] = [445,448],$$

and

$$[430,438] + [19,20] = [449,458].$$

Adding $[446,478]$ to, successively,

$$\{9\},\ [39,42],\ [73,77],\ [109,111], \\ [113,134],\ [159,178],\ \{180\}$$

produces overlapping intervals covering $[455,658]$. Finally,

$$496 + [159,178] = [655,674].$$

Together these intervals prove the third inclusion in (**??**).

For the fourth inclusion in (**??**), namely $[645,862] \subseteq 3H + 2L$, use

$$3H \supseteq [639,734] \cup \{744\}, \qquad 2L \supseteq \{6\} \cup [70,74] \cup [110,118] \cup \{120\}.$$

The four interval sums

$$[645,740], \quad [709,808], \quad [759,854], \quad [854,862]$$

cover $[645,862]$.

For the fifth inclusion in (**??**), namely $[855,1046] \subseteq 4H + L$, use

$$4H \supseteq [852,982] \cup [984,986], \qquad \{3,57,58,60\} \subseteq L.$$

The intervals

$$[855,985], \quad [912,1042], \quad [1041,1044], \quad [1044,1046]$$

then cover $[855,1046]$. □

Since the five intervals in Lemma **??** overlap, they give

$$[215,1046] \subseteq 5\big(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big).$$

Proposition 12. *Every integer $n \geq 215$ is a sum of exactly five elements of $\big(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)$. Consequently,*

$$[212,\infty) \subseteq F_5(\big(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)).$$

The lower endpoint is sharp: none of the integers 209, 210, and 211 belongs to $F_5(\big(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big))$.

Proof. Lemma **??** gives the stronger inclusion $[215, 1046] \subseteq 5\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$; only the initial interval $[215, 867]$ is needed as the base case for strong induction on n. Let $n \geq 868$, and assume that every integer in $[215, n-1]$ lies in $5\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$. Choose the unique $\rho \in \{5, 6, 7, 8\}$ such that $\rho \equiv n \pmod 4$. Then

$$M = \frac{n-\rho}{4} \geq \frac{868-8}{4} = 215.$$

Since $M < n$, the induction hypothesis gives

$$M = q_1 + \cdots + q_5, \qquad q_i \in \left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right).$$

Because $5 \leq \rho \leq 8$, we may choose $\eta_i \in \{1, 2\}$ such that

$$\eta_1 + \cdots + \eta_5 = \rho.$$

Therefore

$$n = 4M + \rho = \sum_{i=1}^{5} (4q_i + \eta_i).$$

Corollary **??** shows that every summand $4q_i + \eta_i$ belongs to $\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$. Thus $n \in 5\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$, completing the induction.

Finally,

$$\begin{aligned} 212 &= 53 + 53 + 53 + 53, \\ 213 &= 53 + 53 + 53 + 54, \\ 214 &= 53 + 53 + 54 + 54, \end{aligned}$$

so $[212, \infty) \subseteq F_5(\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right))$. We next verify that the only elements of $\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$ below 212 are the eight members of L. Primitive Dyck words of lengths 4, 6, and 8 give, after division by 4, respectively,

$$\{3\}, \qquad \{13, 14\}, \qquad \{53, 54, 57, 58, 60\}.$$

Every primitive Dyck word of length 10 has the form $1u0$, where u is a Dyck word of length 8. The lexicographically least such word is 1101010100, whose value is 852; after division by 4 this gives 213. Longer primitive words have still larger values. Hence, with L as in (**??**),

$$\left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right) \cap [0, 211] = L.$$

Let

$$D = \{53, 54, 57, 58, 60\} \subseteq L.$$

A sum of at most five members of L using at most three terms from D is at most

$$3 \cdot 60 + 2 \cdot 14 = 208,$$

whereas a sum using at least four terms from D is at least

$$4 \cdot 53 = 212.$$

Hence $209, 210, 211 \notin F_5(\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right))$. $\square$

5 The complete exceptional set

We first translate six-summand representability in the congruence class 2 modulo 4 into a finite-sum condition on $\left(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD},\geq 12}\right)$.

Lemma 13. *Let $N = 4m+2$. Then N is a sum of at most six positive primitive Dyck numbers if and only if*

$$m \in F_5(\left(\tfrac{1}{4}\cdot\mathcal{N}_{\mathrm{pD},\geq 12}\right)) \cup \left(1 + F_3(\left(\tfrac{1}{4}\cdot\mathcal{N}_{\mathrm{pD},\geq 12}\right))\right) \cup \left(2 + F_1(\left(\tfrac{1}{4}\cdot\mathcal{N}_{\mathrm{pD},\geq 12}\right))\right).$$

Proof. By Lemma **??**, every primitive Dyck summand other than 2 is a multiple of 4. Hence a representation of $N \equiv 2 \pmod 4$ contains an odd number of copies of 2. If it has at most six summands, that number is 1, 3, or 5.

With one copy of 2, division of the remaining sum by 4 gives $m \in F_5(\left(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD},\geq 12}\right))$. With three copies it gives $m-1 \in F_3(\left(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD},\geq 12}\right))$, and with five copies it gives $m-2 \in F_1(\left(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD},\geq 12}\right))$. This proves necessity. Conversely, each of the three memberships gives a representation of N after multiplying the corresponding $\left(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD},\geq 12}\right)$-summands by 4 and adjoining, respectively, one, three, or five copies of 2. Thus the condition is also sufficient. □

Lemma 14 (Small multiples of four)**.** *Among the positive multiples of* 4 *below* 100*, the only integer not representable by at most six positive primitive Dyck numbers is* 44*. Moreover,*

$$44 = 12 + 12 + 12 + 2 + 2 + 2 + 2.$$

Proof. The positive primitive Dyck numbers below 100 are precisely

$$2,\ 12,\ 52,\ 56.$$

First suppose $N < 52$ and write $N = 4x$. A sum of copies of 12 and pairs $2+2$ has the form

$$x = 3a + c$$

and uses $a + 2c$ summands. Under the condition $a + 2c \leq 6$, the possible positive values of $x < 13$ are

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12;$$

the sole missing value is $x = 11$, corresponding to $N = 44$.

For $52 \leq N < 100$, one first uses either $52 = 4\cdot 13$ or $56 = 4\cdot 14$. With at most five remaining summands, the residual quotients obtainable from 12 and pairs $2+2$ are

$$R = \{3a + c : a, c \geq 0,\ a + 2c \leq 5\} = \{0, 1, 2, 3, 4, 5, 6, 7, 9, 10, 12, 15\}.$$

A direct inspection of these twelve integers gives

$$[13, 24] \subseteq (13 + R) \cup (14 + R),$$

so every multiple of 4 from 52 through 96 has the required representation. The displayed seven-term representation of 44 finishes the proof. □

Lemma 15 (Finite exceptional-set certificate)**.** *Let*

$$L = \{3, 13, 14, 53, 54, 57, 58, 60\}.$$

Then

$$F_5(L) \cap [0, 211] = \{0, 3, 6, 9\} \cup [12, 17] \cup [19, 20] \cup [22, 23] \\ \cup [25, 37] \cup [39, 48] \cup [52, 208].$$

Moreover,

$$[0, 211] \cap \Big(F_5(L) \cup (1 + F_3(L)) \cup (2 + F_1(L))\Big) \\ = [0, 7] \cup [9, 10] \cup [12, 23] \cup [25, 37] \\ \cup [39, 48] \cup [52, 208].$$

Consequently, the complement in $[0, 211]$ *is*

$$\{8, 11, 24, 38, 49, 50, 51, 209, 210, 211\}.$$

Proof. Starting with $0L = \{0\}$ and using the recursion $jL = (j-1)L + L$, successive addition of the eight elements of L gives the following table, truncated at 211:

j	$F_j(L) \cap [0, 211]$
1	$\{0, 3\} \cup [13, 14] \cup [53, 54] \cup [57, 58] \cup \{60\}$
2	$\{0, 3, 6\} \cup [13, 14] \cup [16, 17] \cup [26, 28] \cup [53, 54] \cup [56, 58]$
	$\cup [60, 61] \cup \{63\} \cup [66, 68] \cup [70, 74] \cup [106, 108] \cup [110, 118] \cup \{120\}$
3	$\{0, 3, 6, 9\} \cup [13, 14] \cup [16, 17] \cup [19, 20] \cup [26, 31] \cup [39, 42] \cup [53, 54]$
	$\cup [56, 61] \cup [63, 64] \cup [66, 77] \cup [79, 88] \cup [106, 134] \cup [159, 178] \cup \{180\}$
4	$\{0, 3, 6, 9\} \cup [12, 14] \cup [16, 17] \cup [19, 20] \cup [22, 23] \cup [26, 34] \cup [39, 45]$
	$\cup [52, 64] \cup [66, 102] \cup [106, 148] \cup [159, 194]$
5	$\{0, 3, 6, 9\} \cup [12, 17] \cup [19, 20] \cup [22, 23] \cup [25, 37] \cup [39, 48] \cup [52, 208]$.

Also,

$$(1 + F_3(L)) \cap [0, 211] = \{1, 4, 7, 10\} \cup [14, 15] \cup [17, 18] \cup [20, 21] \\ \cup [27, 32] \cup [40, 43] \cup [54, 55] \cup [57, 62] \\ \cup [64, 65] \cup [67, 78] \cup [80, 89] \cup [107, 135] \\ \cup [160, 179] \cup \{181\},$$

and

$$(2 + F_1(L)) \cap [0, 211] = \{2, 5\} \cup [15, 16] \cup [55, 56] \cup [59, 60] \cup \{62\}.$$

Taking the union of these displayed interval lists gives the second formula, and its omitted integers are exactly those in the final set. Each row of the table is obtained from the preceding one by translating by the eight elements of L and taking the union with the preceding row. ☐

Proof of Theorem **??**. Suppose first that $N \equiv 0 \pmod 4$. If $N \geq 100$, write $N = 4n$. Then $n \geq 25$, so Proposition **??** and (**??**) give a representation of N with at most six primitive Dyck summands. Lemma **??** treats the remaining positive multiples of 4 and shows that $r(44) = 7$.

Now let $N \equiv 2 \pmod 4$, say $N = 4m + 2$. If $N \geq 850$, then $m \geq 212$, so Proposition **??** gives $m \in F_5\big(\big(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD},\geq 12}\big)\big)$. By Lemma **??**, N is a sum of at most six primitive Dyck numbers. Together with the preceding paragraph, this already shows that every even $N \geq 848$ has such a representation.

It remains to classify the numbers $N \leq 846$ with $N \equiv 2 \pmod 4$. Here $0 \leq m \leq 211$, and Proposition **??** shows that the only elements of $\big(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD},\geq 12}\big)$ not exceeding m lie in

$$L = \{3, 13, 14, 53, 54, 57, 58, 60\}.$$

Lemma **??** shows that the values of m not covered by the criterion are

$$\{8, 11, 24, 38, 49, 50, 51, 209, 210, 211\}.$$

By Lemma **??**, the corresponding values $N = 4m + 2$ are precisely

$$34, 46, 98, 154, 198, 202, 206, 838, 842, 846.$$

Thus these ten numbers, together with 44, are exactly the positive even integers that are not sums of at most six primitive Dyck numbers.

All of them except 46 have seven-summand representations:

$$\begin{aligned}
34 &= 12 + 12 + 2 + 2 + 2 + 2 + 2,\\
44 &= 12 + 12 + 12 + 2 + 2 + 2 + 2,\\
98 &= 56 + 12 + 12 + 12 + 2 + 2 + 2,\\
154 &= 52 + 52 + 12 + 12 + 12 + 12 + 2,\\
198 &= 56 + 52 + 52 + 12 + 12 + 12 + 2,\\
202 &= 56 + 56 + 52 + 12 + 12 + 12 + 2,\\
206 &= 56 + 56 + 56 + 12 + 12 + 12 + 2,\\
838 &= 240 + 240 + 240 + 52 + 52 + 12 + 2,\\
842 &= 240 + 240 + 240 + 56 + 52 + 12 + 2,\\
846 &= 240 + 240 + 240 + 56 + 56 + 12 + 2.
\end{aligned}$$

Proposition **??** shows that 46 requires exactly eight summands. This proves all assertions of the theorem. □

Corollary 16. *The integer* 848 *is the least even integer* T *such that every even integer* $N \geq T$ *is a sum of at most six primitive Dyck numbers.*

Proof. Theorem **??** shows that six summands suffice for every even integer at least 848, whereas 846 requires seven summands. □

Corollary 17. *Every nonnegative even integer is a sum of at most eight elements of* $\mathcal{N}_{\mathrm{pD}}$*, and* 46 *is the unique even integer that is not a sum of at most seven elements of* $\mathcal{N}_{\mathrm{pD}}$*.*

Proof. The assertion for positive integers follows immediately from Theorem **??**; the integer 0 is the empty sum. □

For $A \subseteq 2\mathbb{N}_0$, we call A a *relative additive basis of order at most h for* $2\mathbb{N}_0$ if every element of $2\mathbb{N}_0$ is a sum of at most h elements of A. Its relative order is *exactly* h if h is the least integer with this property. We similarly call A a *relative asymptotic additive basis of order at most h* if every sufficiently large element of $2\mathbb{N}_0$ is a sum of at most h elements of A.

Theorem **??** gives the complete exceptional set for six-summand representability. Corollary **??** shows that $\mathcal{N}_{\mathrm{pD}}$ is a relative additive basis of exact order eight for $2\mathbb{N}_0$, while Corollary **??** identifies the sharp six-summand threshold. The next section proves that the relative asymptotic order is exactly six.

6 Recurring gaps and asymptotic optimality

Representations of an integer congruent to 2 modulo 4 split according to the number of copies of the exceptional summand 2. For five summands the criterion is as follows.

Lemma 18 (Five-summand criterion). *Let $N = 4m + 2$ with $m \geq 3$. Then $r(N) \leq 5$ if and only if*

$$m \in F_4\big(\big(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)\big) \cup \big(1 + F_2\big(\big(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)\big)\big). \tag{23}$$

Proof. By Lemma **??**, a representation of N uses some number $s \geq 0$ of copies of 2 and otherwise uses summands from $4\big(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)$. Reduction modulo 4 gives $2s \equiv 2 \pmod 4$, so s is odd. If at most five summands are used, then $s \in \{1, 3, 5\}$. Writing the remaining summands as $4p_i$ and dividing $N = 2s + \sum_i 4p_i$ by 4 after subtracting 2 gives

$$m = \frac{s-1}{2} + \sum_i p_i.$$

For $s = 1$ this gives $m \in F_4\big(\big(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)\big)$, and for $s = 3$ it gives $m \in 1 + F_2\big(\big(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)\big)$. If $s = 5$, then $m = 2$, excluded by $m \geq 3$. This proves necessity. Conversely, membership in the first set of (**??**) gives a representation using one copy of 2, and membership in the second gives one using three copies of 2. $\square$

The following lemma isolates the lower-bound mechanism.

Lemma 19 (Recurring obstruction). *For every integer $k \geq 2$, put*

$$x_k = 10 \cdot 4^k - 2.$$

Then

$$x_k \notin F_4\big(\big(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)\big) \qquad \textit{and} \qquad x_k - 1 \notin F_2\big(\big(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)\big). \tag{24}$$

Proof. Fix $k \geq 2$, and abbreviate $x = x_k$. The least element of generation $k + 2$ is

$$g_{k+2} > 3 \cdot 4^{k+1} = 12 \cdot 4^k > x.$$

Consequently, every element of $\big(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)$ that is at most x has generation at most $k + 1$. By Corollary **??**, every such element is either *small*, of generation at most k and hence at most

$$U_k = 4^k - 2^{k-1},$$

or *large*, of generation $k+1$ and hence between g_{k+1} and $U_{k+1} = 4^{k+1} - 2^k$.

First consider $x - 1 = 10 \cdot 4^k - 3$. It is larger than U_{k+1} and smaller than g_{k+2}, so it does not itself belong to $\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$. Moreover, any sum of two eligible elements is at most

$$2U_{k+1} = 8 \cdot 4^k - 2^{k+1} < 10 \cdot 4^k - 3 = x - 1.$$

Thus $x - 1 \notin F_2(\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right))$.

Now suppose, for contradiction, that

$$x = q_1 + \cdots + q_t, \qquad t \leq 4, \qquad q_i \in \left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right).$$

Let λ be the number of large summands. If $\lambda \leq 2$, then filling the remaining positions with the largest possible small summands gives

$$\begin{aligned} x &\leq 2U_{k+1} + 2U_k \\ &= 2(4^{k+1} - 2^k) + 2(4^k - 2^{k-1}) \\ &= 10 \cdot 4^k - 3 \cdot 2^k < 10 \cdot 4^k - 2 = x, \end{aligned}$$

a contradiction. Hence $\lambda \geq 3$. But then the three large summands alone have sum at least

$$3g_{k+1} = 10 \cdot 4^k - 1 = x + 1$$

by (**??**), again a contradiction. Therefore $x \notin F_4(\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right))$. □

Proof of Theorem **??**. Fix $k \geq 2$ and put $m = x_k = 10 \cdot 4^k - 2$. Then

$$N_k = 4m + 2 = 10 \cdot 4^{k+1} - 6.$$

By Lemma **??**,

$$m \notin F_4(\left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)), \qquad m - 1 \notin F_2(\left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)).$$

The five-summand criterion, Lemma **??**, therefore gives $r(N_k) \geq 6$. The integer N_k is not one of the eleven exceptions in Theorem **??**, so $r(N_k) \leq 6$. Hence $r(N_k) = 6$. Since $N_k \to \infty$, no $h \leq 5$ bounds $r(N)$ for all sufficiently large even N, while Theorem **??** gives the upper bound $h = 6$. Therefore $h_{\mathrm{asym}} = 6$. □

Corollary 20 (Order of $\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$)**.** *The set $\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)$ is an asymptotic additive basis of order exactly* 5 *for the nonnegative integers. More precisely, every integer $n \geq 212$ lies in $F_5(\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right))$, whereas $F_4(\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right))$ and $F_3(\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right))$ each omit infinitely many integers.*

Proof. Proposition **??** gives $[212, \infty) \subseteq F_5(\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right))$. For every $k \geq 2$, Lemma **??** gives

$$x_k = 10 \cdot 4^k - 2 \notin F_4(\left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)).$$

Since $x_k \to \infty$, four summands do not suffice asymptotically. The same is true for three summands because $F_3(\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right)) \subseteq F_4(\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\right))$. □

Corollary 21 (Multiples of 4 requiring five summands). *For every integer $k \geq 3$,*

$$M_k = 10 \cdot 4^{k+1} - 8$$

satisfies $r(M_k) = 5$. Hence infinitely many multiples of 4 require exactly five positive primitive Dyck summands.

Proof. Write $M_k = 4x_k$. Since $k \geq 3$, one has $x_k \geq 212$, and Proposition **??** gives $x_k \in F_5\big(\big(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)\big)$. Multiplying such a representation by 4 shows that $r(M_k) \leq 5$.

Suppose that M_k had a representation using at most four primitive Dyck summands, and let s be the number of copies of 2. Since $M_k \equiv 0 \pmod 4$, s is even, so $s \in \{0, 2, 4\}$. If $s = 0$, division by 4 gives $x_k \in F_4\big(\big(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)\big)$, contradicting Lemma **??**. If $s = 2$, division after removing the two copies of 2 gives $x_k - 1 \in F_2\big(\big(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)\big)$, again a contradiction. If $s = 4$, no other summand is available and division by 4 would give $x_k = 2$, which is impossible. Thus $r(M_k) \geq 5$, and equality follows. □

Remark 22 (The two residue classes). If $N = 4n$, then $n \in F_5\big(\big(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)\big)$ for all sufficiently large n, so $r(N) \leq 5$ eventually; Corollary **??** shows that equality occurs infinitely often. If $N = 4m + 2$, the number of copies of 2 must be odd. For $m = x_k$, the one-copy case would require $m \in F_4\big(\big(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)\big)$ and the three-copy case would require $m - 1 \in F_2\big(\big(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\geq 12}\big)\big)$; Lemma **??** excludes both. This residue class raises the relative asymptotic order from 5 to 6.

7 OEIS sequences and computational verification

Three OEIS sequences occur naturally. The totally balanced numbers form A014486 [?]; the positive primitive Dyck numbers form A057547 [?]; and the central sequence of this paper is

$$a(n) = r(2n),$$

namely A395858 [?]. Theorem **??** determines all terms exceeding six:

$$a(n) = 8 \iff n = 23,$$

and

$$a(n) = 7 \iff n \in \{17, 22, 49, 77, 99, 101, 103, 419, 421, 423\}.$$

Theorem **??** supplies an explicit infinite subsequence at level six:

$$a(5 \cdot 4^{k+1} - 3) = 6 \qquad (k \geq 2). \tag{25}$$

The program `A395858.py` at commit `3525cfb` reproduces the displayed initial terms and generates the first 50,000 terms of A395858. The supplementary verification script at the same fixed commit checks the five finite interval inclusions, the sharp lower endpoint for Proposition **??**, and the finite exceptional-set certificate. Its independent dynamic-programming computation checks $r(N)$ through the selected finite bound (by default 50,000). The default run also checks the recurring obstruction for $2 \leq k \leq 5$, the six-summand family

$$r(10 \cdot 4^{k+1} - 6) = 6$$

for $2 \le k \le 5$, and the five-summand family

$$r(10 \cdot 4^{k+1} - 8) = 5$$

for $3 \le k \le 5$. This version passed the repository's Python CI check at commit `3525cfb` (Python CI run #1). These computations provide reproducible finite cross-checks; the assertions for all k and all sufficiently large integers are proved theoretically above.

8 Concluding remarks

The results determine three exact orders associated with primitive Dyck numbers. The relative additive order on all nonnegative even integers is 8, the relative asymptotic order is 6, and the asymptotic additive order of $\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD},\ge 12}\right)$ is 5. The same base-4 self-similarity that propagates long five-summand intervals also produces the recurring obstruction $x_k = 10 \cdot 4^k - 2$.

It remains to characterize the complete set of even integers N for which $r(N) = 6$. Computations suggest additional block structure near the points $10 \cdot 4^k$, but the exact widths and endpoints of those blocks are not yet proved. A broader question is whether analogous regular-subfamily upper bounds and generation-gap lower bounds occur for other indecomposable digital languages.

References

[1] J. P. Bell, K. G. Hare, and J. Shallit, When is an automatic set an additive basis?, *Proc. Amer. Math. Soc. Ser. B* **5** (2018), 50–63.

[2] J. P. Bell, T. F. Lidbetter, and J. Shallit, Additive number theory via approximation by regular languages, *Internat. J. Found. Comput. Sci.* **31** (2020), 667–687.

[3] P. Madhusudan, D. Nowotka, A. Rajasekaran, and J. Shallit, Lagrange's theorem for binary squares, in *43rd International Symposium on Mathematical Foundations of Computer Science*, Leibniz Int. Proc. Inform., Vol. 117, Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2018, pp. 18:1–18:14.

[4] M. B. Nathanson, *Additive Number Theory: The Classical Bases*, Grad. Texts in Math., Vol. 164, Springer, 1996.

[5] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences, A014486*, `https://oeis.org/A014486`.

[6] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences, A057547*, `https://oeis.org/A057547`.

[7] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences, A395858*, `https://oeis.org/A395858`.

[8] A. Rajasekaran, J. Shallit, and T. Smith, Sums of palindromes: an approach via automata, in *35th Symposium on Theoretical Aspects of Computer Science*, Leibniz Int. Proc. Inform., Vol. 96, Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2018, pp. 54:1–54:12.

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